

Remaining Gaps & 5-Member Implementation Task Plan

Architectural Audit, Remaining Gap Matrix & Sprint Execution Roadmap

PROJECT METADATA & BASELINE AUDIT SUMMARY

Platform: EdLearn Learning Ecosystem	Document Date: August 20, 2026
Frontend: Next.js 16.2 (App Router), React 19, Tailwind v4	Current Status: Baseline Upgrades Complete (84/84 Tests Passed)
Backend: Express 4.x, TypeScript, Prisma 7.8	Typecheck Health: 0 Errors (Frontend & Backend)
Databases: PostgreSQL 16, MongoDB 7.0, Redis 7.2	Target Release: Production Release Candidate (v2.0)

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1. Executive Summary & Current Baseline Architecture

EdLearn is a full-stack educational platform designed to provide multi-modal AI tutoring, structured learning roadmaps, and content curation. Over the initial engineering sprints, the platform established a robust baseline comprising Next.js 16, React 19, TypeScript, Tailwind CSS v4, Express, PostgreSQL with Prisma 7.8, Redis caching, and multi-provider AI failovers.

CURRENT PLATFORM VERIFICATION STATUS (ALL 84 TESTS PASSING)

- Backend Vitest Suite: 84 / 84 tests passing across 6 test suites (auth, content, gdpr, generate, roadmap, platform_features).
- TypeScript Typecheck: 0 compilation errors across both frontend (npx tsc --noEmit) and backend (npm run typecheck).
- Database Schema: PostgreSQL 16 schema generated with UUID primary keys, compound indexes, and cascade integrity.

2. Completed vs Remaining Features Comparison Matrix

The following table establishes the exact source-of-truth baseline for EdLearn, clearly distinguishing completed components from remaining gaps to prevent duplicate assignments.

Subsystem / Area	Current Implementation State	Status	Remaining Action Required
Authentication & 2FA	JWT access/refresh rotation, TOTP 2FA, backup codes, Google OAuth.	COMPLETED	None. Stable and tested.
Pedagogical Modes 1-7	Accelerator, Socratic, Simplifier, Roadmap, Gap Finder, Feynman, Duo Podcast.	COMPLETED	None. AI prompt engine active.
Content Hubs	Books summaries, Media (YouTube/podcasts), News aggregator with crons.	COMPLETED	None. Curation crons functional.
Bookmarks & Search	Universal bookmarks (/bookmarks) and Cmd+K global search modal.	COMPLETED	None. Full-stack active.
Progress & Streaks	Timezone-aware streak counter and 365-day activity heatmap on dashboard.	COMPLETED	None. Redis cached.
Quizzes & Notes	QuizRunner interactive tester with scoring and personal notes editor.	COMPLETED	None. Real-time auto-saving.
RBAC & Admin Portal	Role middleware (ADMIN, USER, MOD) and telemetry portal at /admin.	COMPLETED	None. Protected route active.
In-App Notifications	NotificationCenter bell with unread badge counter and read toggles.	COMPLETED	None. Polling active.
AI Provider Failover	Automatic circuit breaker switching between Groq and Gemini on 429/5xx.	COMPLETED	None. Fully resilient.
Spaced Repetition (SM-2)	Quiz attempts saved, but incorrect answers not scheduled for recall.	NOT STARTED	Build SM-2 flashcard review engine.
Live Collaborative Rooms	Single-player study only. No real-time peer audio/whiteboard.	NOT STARTED	Implement Socket.io study rooms.
PWA & Offline Sync	Requires active connection. No offline cache or background sync queue.	NOT STARTED	Add Service Worker & IndexedDB.
Topic Discussions & Q&A	Notes are private. No public lesson discussions or peer answers.	NOT STARTED	Add threaded Q&A under topics.
Automated Email Digests	In-app notifications only. No scheduled weekly email reports.	NOT STARTED	Build Nodemailer/Resend cron worker.
Verifiable Certificates	Awards in-app badges only. No downloadable PDF certificates.	NOT STARTED	Build verifiable PDF certificate engine.

3. Detailed Remaining Gap Analysis (10 Core Gaps)

Each remaining gap has been rigorously audited against the codebase to ensure no duplicate work is assigned. The following 10 gaps represent the concrete scope for the 5-member team.

Gap ID	Category	Description & Missing Functionality	Priority	Impact	Owner
GAP-01	Learning Gap	Spaced Repetition System (SM-2): Missed quiz answers are not scheduled into SuperMemo-2 review intervals for long-term retention.	P0	High Learning Retention	Member 2 / 3
GAP-02	Collaboration	Real-Time Peer Study Rooms: No synchronous study sessions, shared Pomodoro timers, or collaborative whiteboard notes.	P0	High Engagement	Member 4 / 1
GAP-03	PWA / Offline	PWA Offline Cache & Sync Queue: Inability to review cached study notes or queue quiz attempts during network dropouts.	P1	Mobile & Offline Access	Member 1 / 5
GAP-04	Community	Threaded Topic Discussions & Q&A: Lack of peer-to-peer lesson discussion threads, code question upvoting, and moderation.	P1	Social Learning	Member 4 / 3
GAP-05	Engagement	Automated Email Digest & Retention Crons: No weekly progress emails, streak reminder alerts, or personalized digest delivery.	P1	User Re-engagement	Member 2 / 5
GAP-06	Credentialing	Cryptographically Verifiable Certificates: Missing official PDF completion certificates with unique SHA-256 validation hashes.	P1	Academic Credibility	Member 2 / 1
GAP-07	Monetization	Subscription Tiers & Stripe Webhooks: Credit consumption is flat; lacks Pro/Student tier gating and automated webhook quota resets.	P2	Platform Scalability	Member 2 / 3
GAP-08	Quality Assur.	Playwright E2E Test Suite: Vitest unit/integration tests pass, but end-to-end browser journeys are unautomated.	P1	Regression Defense	Member 5
GAP-09	AI & Discovery	Semantic Recommendation Engine: Recommendations rely on simple genre filters rather than cosine embedding similarity.	P2	Discovery Relevance	Member 3 / 2
GAP-10	DevOps & Logs	Centralized Log Streaming & Backup Automation: Missing automated pg_dump cron jobs and structured Pino log aggregation.	P1	Disaster Recovery	Member 5

4. Detailed Task Breakdown: Epic -> Feature -> Task -> Subtasks

EPIC 1: Retention & Spaced Repetition (GAP-01)

- **FEATURE 1.1: SM-2 Flashcard Review Engine: Automate conversion of missed quiz questions into daily spaced repetition decks.**
 - Subtask 1.1.1: Add Flashcard and FlashcardReview models to Prisma schema with easeFactor, interval, and repetitions fields.
 - Subtask 1.1.2: Implement SM-2 mathematical algorithm service in backend calculating next review dates (1d, 6d, interval*EF).
 - Subtask 1.1.3: Build POST /api/flashcards/review endpoint to record recall grades (0-5) and update card scheduling.
 - Subtask 1.1.4: Build interactive FlipCard.tsx and /flashcards study runner UI with keyboard navigation (Space to flip, 1-4 to grade).
 - Subtask 1.1.5: Hook quiz failure event to auto-create flashcards for incorrect questions.

EPIC 2: Real-Time Social Learning & Study Rooms (GAP-02, GAP-04)

- **FEATURE 2.1: Collaborative Live Study Rooms: Multi-user virtual study rooms with synchronized Pomodoro timers and shared canvas.**
 - Subtask 2.1.1: Initialize Socket.io server with room namespaces (/study-room/:roomId) and JWT handshake authentication.
 - Subtask 2.1.2: Implement synchronized room state (timer start/pause/reset, active participants list, presence heartbeats).
 - Subtask 2.1.3: Build collaborative shared scratchpad with operational transform / debounced broadcast updates.
 - Subtask 2.1.4: Create frontend /study-rooms/[id] layout with participant avatars, chat dock, and synced timer widget.
- **FEATURE 2.2: Threaded Lesson Discussion Forum: Peer Q&A threads embedded under each roadmap topic lesson.**
 - Subtask 2.2.1: Add TopicComment model in Prisma with self-referential parentId for infinite threading and upvote counts.
 - Subtask 2.2.2: Create GET/POST /api/topics/:topicId/comments and POST /api/comments/:id/upvote endpoints.
 - Subtask 2.2.3: Build TopicDiscussionSection.tsx with markdown rendering, reply nesting, and moderator delete badges.

EPIC 3: Platform Credibility & Certification (GAP-06)

- **FEATURE 3.1: Verifiable Course Completion Certificates: Generate tamper-proof PDF certificates with public verification.**
 - Subtask 3.1.1: Add Certificate model in Prisma storing certificateId, userId, roadmapId, sha256Hash, and issuedAt.
 - Subtask 3.1.2: Build backend PDF generation service rendering gold-bordered diploma with student name, course title, and QR code.
 - Subtask 3.1.3: Create public verification route GET /api/certificates/verify/:hash and frontend /verify/[hash] lookup page.

5. Member 1 Task Sheet: Frontend, UX & Interactive Learning

OWNER: MEMBER 1

MEMBER 1 WORKSTREAM PROFILE

- Primary Role: Lead Frontend Engineer & UX Architect
- Git Branch: feature/member1-frontend-ux
- Core Ownership: Flashcards study UI, PWA service worker, collaborative study room interface, certificate viewer.
- Shared Touchpoints: frontend/src/components/Document/LivingDocument.tsx, frontend/src/components/Layout/Sidebar.tsx

Assigned Tasks & Subtasks:

- **TASK M1-01 (GAP-01): Interactive Flashcards Study Canvas (/flashcards)**
 - Build 3D card-flip component with smooth CSS perspective transitions.
 - Implement 1-5 recall rating buttons (Again, Hard, Good, Easy) with keyboard shortcuts (1, 2, 3, 4, Space).
 - Display daily due cards counter, deck completion summary, and streak celebration confetti.
- **TASK M1-02 (GAP-02): Virtual Study Room UI & Shared Canvas (/study-rooms/[id])**
 - Build responsive study room layout with synced circular Pomodoro timer widget.
 - Create live participant strip showing user avatars, audio pulse indicators, and active study badges.
 - Implement real-time room chat drawer and shared markdown scratchpad with live typing indicators.
- **TASK M1-03 (GAP-03): PWA Service Worker & Offline Storage**
 - Configure next-pwa / Workbox service worker caching for static assets and generated notes.
 - Build offline indicator banner and IndexedDB queueing for offline quiz submissions.
- **TASK M1-04 (GAP-06): Certificate Modal & Verification Page (/verify/[hash])**
 - Create celebratory certificate unlock modal on roadmap completion.
 - Build responsive public verification page displaying certificate authenticity badge, student name, and course details.

Task ID	Deliverable Module	Est. Hours	Dependencies	Acceptance Gate
M1-01	frontend/src/app/flashcards/page.tsx	14h	M3-01 (DB), M2-01 (API)	Flashcards flip & grade smoothly.
M1-02	frontend/src/app/study-rooms/[id]/page.tsx	18h	M4-01 (Socket.io)	Synced timer & live peer presence.
M1-03	frontend/public/sw.js, IndexedDB sync	12h	None	Notes load offline without network.
M1-04	frontend/src/app/verify/[hash]/page.tsx	8h	M2-03 (Certificate API)	Public certificate verification renders.

6. Member 2 Task Sheet: Backend, APIs & Business Logic

OWNER: MEMBER 2

MEMBER 2 WORKSTREAM PROFILE

- Primary Role: Lead Backend Engineer & Systems Architect
- Git Branch: feature/member2-backend-apis
- Core Ownership: SM-2 algorithm engine, email digest cron workers, certificate generation & validation, Stripe billing.
- Shared Touchpoints: backend/src/index.ts, backend/src/services/contentCrons.ts

Assigned Tasks & Subtasks:

- **TASK M2-01 (GAP-01): Spaced Repetition (SM-2) Scheduling Engine**
 - Implement sm2Service.ts: calculate new interval, repetitions, and easeFactor based on user quality rating (0-5).
 - Build GET /api/flashcards/due (returns cards due today) and POST /api/flashcards/review.
 - Add automatic card creation hook when QuizAttempt score is below 80%.
- **TASK M2-02 (GAP-05): Automated Email Digest & Retention Worker**
 - Build emailService.ts using Nodemailer / Resend with HTML email templates.
 - Implement weeklyDigestCron.ts running every Sunday at 08:00 UTC sending streak stats, weekly hours, and next lessons.
 - Add user email notification preferences endpoint (PATCH /api/user/preferences/email).
- **TASK M2-03 (GAP-06): Certificate Generation & Cryptographic Verification**
 - Implement certificateService.ts using PDFKit / jsPDF generating high-res diploma with SHA-256 signature.
 - Build POST /api/certificates/generate/:roadmapId and GET /api/certificates/verify/:hash.
- **TASK M2-04 (GAP-07): Subscription Tiers & Stripe Webhook Handler**
 - Build stripeService.ts for Pro/Student subscription checkout sessions.
 - Implement POST /api/billing/webhook with signature verification handling invoice.paid and customer.subscription.deleted.

Task ID	Deliverable Module	Est. Hours	Dependencies	Acceptance Gate
M2-01	backend/src/services/sm2Service.ts	14h	M3-01 (Schema)	SM-2 intervals compute accurately.
M2-02	backend/src/services/emailDigest.ts	12h	M3-02 (User Prefs)	Digest emails deliver on schedule.
M2-03	backend/src/services/certificate.ts	10h	M3-03 (Cert Model)	SHA-256 signature verified.
M2-04	backend/src/routes/billing.ts	14h	M3-04 (Sub Model)	Stripe webhooks update user tiers.

7. Member 3 Task Sheet: Database, Learning System & Core Data

OWNER: MEMBER 3

MEMBER 3 WORKSTREAM PROFILE

- Primary Role: Lead Database Engineer & Data Architect
- Git Branch: feature/member3-database-core
- Core Ownership: Prisma schema extensions, database migrations, indexes, vector similarity queries, data integrity.
- Shared Touchpoints: backend/prisma/schema.prisma, backend/src/lib/db.ts, backend/src/tests/setup.ts

Assigned Tasks & Subtasks:

- **TASK M3-01 (GAP-01, GAP-04): Prisma Schema Extensions & Migrations**
 - Add Flashcard model: id, userId, dayId, front, back, easeFactor (Float @default(2.5)), interval (Int @default(0)), repetitions (Int @default(0)), dueDate (DateTime).
 - Add TopicComment model: id, userId, topicId, parentId (nullable), content, upvotes (Int @default(0)), createdAt.
 - Add Certificate model: id, userId, roadmapId, sha256Hash (unique), pdfUrl, issuedAt.
 - Run npx prisma migrate dev and update mockDb in backend/src/tests/setup.ts.
- **TASK M3-02 (GAP-04): Discussion Threads & Moderation Data Layer**
 - Build database queries for nested hierarchical comment trees with pagination.
 - Implement database transactions for atomic comment upvoting and preventing duplicate user upvotes.
- **TASK M3-03 (GAP-09): Semantic Vector Similarity & Tag Recommendation Engine**
 - Implement tag and topic similarity query in recommendationService.ts matching student difficulty and study history.
 - Add composite indexes on Topic(dayId, title) and Progress(userId, completedAt) for query acceleration.

Task ID	Deliverable Module	Est. Hours	Dependencies	Acceptance Gate
M3-01	backend/prisma/schema.prisma	12h	None (Foundation)	Prisma migration clean, 0 orphans.
M3-02	backend/src/routes/comments.ts	12h	M3-01 (Schema)	Nested comments query < 50ms.
M3-03	backend/src/services/recommendation.ts	14h	M3-01 (Indexes)	Personalized recommendations return.
M3-04	backend/src/tests/setup.ts	6h	M3-01 (Schema)	All mockDb methods up-to-date.

8. Member 4 Task Sheet: Advanced Features & Community

OWNER: MEMBER 4

MEMBER 4 WORKSTREAM PROFILE

- Primary Role: Lead Real-Time Systems & Community Engineer
- Git Branch: feature/member4-collaboration-community
- Core Ownership: Socket.io real-time server, WebRTC signaling, live study room sync, topic Q&A discussion engine.
- Shared Touchpoints: backend/src/index.ts, frontend/src/components/Document/LivingDocument.tsx

Assigned Tasks & Subtasks:

• TASK M4-01 (GAP-02): Socket.io Real-Time Study Room Server

- Set up Socket.io server with Redis adapter for multi-instance horizontal scaling.
- Implement room lifecycle events: join-room, leave-room, timer-sync, chat-message, scratchpad-delta.
- Add heartbeat ping/pong mechanism to prune disconnected participants within 10s.

• TASK M4-02 (GAP-02): Collaborative Whiteboard / Scratchpad Sync

- Build operational broadcast protocol for shared study notes in active study rooms.
- Implement state conflict resolution so simultaneous typers do not overwrite each other.

• TASK M4-03 (GAP-04): Threaded Topic Discussion & Q&A Service

- Build backend controller for topic discussions: GET /api/topics/:topicId/comments, POST comments, POST upvote.
- Implement content moderation filter screening vulgarity and auto-flagging spam.

Task ID	Deliverable Module	Est. Hours	Dependencies	Acceptance Gate
M4-01	backend/src/lib/socket.ts	16h	M3-01 (DB)	Rooms support 50+ concurrent users.
M4-02	backend/src/services/roomSync.ts	14h	M4-01 (Socket)	Scratchpad syncs < 100ms latency.
M4-03	backend/src/routes/comments.ts	12h	M3-02 (DB Query)	Threaded replies & upvotes active.

9. Member 5 Task Sheet: Testing, DevOps & Production QA

OWNER: MEMBER 5

MEMBER 5 WORKSTREAM PROFILE

- Primary Role: Lead QA, Security & DevOps Engineer
- Git Branch: feature/member5-quality-devops
- Core Ownership: Playwright E2E testing, Docker multi-stage hardening, database backup crons, Prometheus alerts, security audit.
- Shared Touchpoints: backend/Dockerfile, docker-compose.yml, .github/workflows/ci.yml

Assigned Tasks & Subtasks:

- **TASK M5-01 (GAP-08): Playwright End-to-End Test Suite**
 - Configure Playwright with test fixtures for authenticated student sessions.
 - Implement E2E specs: e2e/auth-flow.spec.ts, e2e/study-flow.spec.ts (create roadmap -> study notes -> quiz -> pass -> streak update), e2e/admin-flow.spec.ts.
 - Integrate Playwright runs into GitHub Actions CI pipeline.
- **TASK M5-02 (GAP-10): Production Docker Hardening & Database Backup Automation**
 - Optimize backend and frontend multi-stage Dockerfiles with unprivileged non-root users and slim Node 20 base.
 - Build automated PostgreSQL daily backup script (scripts/backup_db.sh) with S3 upload and retention pruning.
 - Configure Prometheus alerts for 5xx spike (>1%), Redis memory (>80%), and DB connection pool saturation.
- **TASK M5-03 (GAP-08, GAP-10): OWASP Security & Dependency Audit**
 - Audit JWT token validation, rate limiters, and CORS configurations.
 - Run npm audit and automated vulnerability scans across all backend and frontend dependencies.

Task ID	Deliverable Module	Est. Hours	Dependencies	Acceptance Gate
M5-01	frontend/e2e/*.spec.ts, CI workflow	16h	All Feature Streams	E2E tests pass in headless CI.
M5-02	Dockerfile, scripts/backup_db.sh	12h	None	Docker image < 200MB, backups run.
M5-03	Prometheus alerts & OWASP audit	10h	None	0 critical/high vulnerabilities.

10. Cross-Functional Responsibility & Ownership Matrix

To guarantee zero merge conflicts and absolute accountability, every remaining task has exactly ONE Primary Owner (P), with designated Reviewers (R), Dependencies (D), and Support (S).

Task ID & Description	Member 1 (UI)	Member 2 (API)	Member 3 (DB)	Member 4 (Realtime)	Member 5 (QA/DevOps)
M1-01: Flashcards Study UI	PRIMARY (P)	Dependency (D)	Dependency (D)	—	Review (R)
M1-02: Study Room Layout	PRIMARY (P)	—	—	Dependency (D)	Review (R)
M1-03: PWA Offline Caching	PRIMARY (P)	—	—	—	Support (S)
M2-01: SM-2 Algorithm Engine	Review (R)	PRIMARY (P)	Dependency (D)	—	Review (R)
M2-02: Email Digest Worker	—	PRIMARY (P)	Dependency (D)	—	Support (S)
M2-03: Certificate Generation	Review (R)	PRIMARY (P)	Dependency (D)	—	Review (R)
M3-01: Prisma Schema & Migrations	—	Dependency (D)	PRIMARY (P)	Dependency (D)	Review (R)
M3-03: Vector Recommendations	Review (R)	Dependency (D)	PRIMARY (P)	—	Review (R)
M4-01: Socket.io Room Server	Dependency (D)	—	Dependency (D)	PRIMARY (P)	Review (R)
M4-03: Threaded Topic Q&A	Dependency (D)	—	Dependency (D)	PRIMARY (P)	Review (R)
M5-01: Playwright E2E Suite	Support (S)	Support (S)	Support (S)	Support (S)	PRIMARY (P)
M5-02: Docker & Backup Automation	—	—	Dependency (D)	—	PRIMARY (P)

CONFLICT PREVENTION & WORKSTREAM BOUNDARY RULES

- Database migrations (Member 3) MUST be merged first before API or UI work begins.
- Members 1 and 4 work in separate route directories (/flashcards, /study-rooms, /verify) to prevent file contention.
- Shared stores (workspaceStore.ts) require PR approval from the Lead Architect before modifying.

11. Dependency Map & Sequential Execution Phases

The implementation of the remaining 10 gaps follows a strict 5-phase dependency order:

PHASE 1: Core Schema & Infrastructure Foundations (Days 1–5)

- Deliverables: Prisma schema extensions (Flashcard, TopicComment, Certificate, SubscriptionTier), database migrations, mockDb updates, and Socket.io server skeleton.
- Primary Owners: Member 3 (Database Lead) & Member 4 (Real-time Lead).

PHASE 2: Backend Business Logic & Algorithmic Engines (Days 6–12)

- Deliverables: SM-2 spaced repetition scheduler, certificate generation service, email digest worker, and Socket.io room lifecycle event handlers.
- Primary Owners: Member 2 (Backend Lead) & Member 4 (Real-time Lead).

PHASE 3: Frontend Interfaces & Interactive Canvases (Days 13–20)

- Deliverables: Interactive flashcards study UI, virtual study room canvas with synced timer, PWA service worker, and topic discussion section.
- Primary Owners: Member 1 (Frontend Lead) & Member 4 (Community Lead).

PHASE 4: End-to-End Integration & Security Hardening (Days 21–25)

- Deliverables: Playwright E2E test suite covering complete user journeys, OWASP dependency audit, and Stripe webhook integration.
- Primary Owners: Member 5 (QA/DevOps Lead) & Member 2 (Backend Lead).

PHASE 5: Production Deployment & Observability (Days 26–28)

- Deliverables: Multi-stage Docker hardening, automated database backup cron verification, Prometheus alert rule configuration, and release sign-off.
- Primary Owners: Member 5 (DevOps Lead) & Lead Architect.

12. 4-Week Sprint Plan (Weekly Execution Roadmap)

Week	Member 1 (UI)	Member 2 (API)	Member 3 (DB)	Member 4 (Realtime)	Member 5 (QA/DevOps)
Week 1	PWA Service Worker setup & UI wireframes.	SM-2 algorithm design & email templates.	Prisma schema models & migrations (M3-01).	Socket.io server skeleton & Redis adapter.	Playwright test harness setup & CI config.
Week 2	Flashcards UI (/flashcards) & card-flip.	SM-2 endpoints & Email digest cron worker.	TopicComment queries & recommendation engine.	Study room event sync & presence tracking.	Auth & study journey E2E specs.
Week 3	Study room canvas & synced timer widget.	Certificate generator & QR verification.	Database index optimization & vacuuming.	Threaded topic Q&A & moderation API.	Docker multi-stage build & backup scripts.
Week 4	Certificate viewer & PWA offline sync.	Stripe subscription webhook handler.	Data migration dry-run & schema lock.	Whiteboard conflict resolution & load test.	Full E2E regression run & security sign-off.

WEEKLY SPRINT GATES & MILESTONES

- End of Week 1: Database migrations merged; all local dev environments updated with new Prisma client.
- End of Week 2: SM-2 and Socket.io endpoints active; flashcard UI connected to backend.
- End of Week 3: Study rooms and certificates operational; E2E tests executing in CI.
- End of Week 4: Zero regressions, 100% test pass rate, Docker images hardened for production release.

13. Git Strategy, Pull Request Protocol & Quality Assurance

13.1 Branching Convention & Isolation

Each member operates on an isolated feature branch branched directly from main:

- **Member 1:** feature/member1-frontend-ux
- **Member 2:** feature/member2-backend-apis
- **Member 3:** feature/member3-database-core
- **Member 4:** feature/member4-collaboration-community
- **Member 5:** feature/member5-quality-devops

13.2 Commit Message Standards (Conventional Commits)

All commit messages must adhere to the Conventional Commits format:

- feat(flashcards): implement 3D card-flip interaction with keyboard shortcuts
- fix(sm2): correct interval calculation when quality rating is 3
- test(e2e): add Playwright spec for complete roadmap study and quiz journey
- chore(docker): add non-root user and optimize layer caching in Dockerfile

13.3 Pull Request Acceptance Gate Checklist

Every pull request must fulfill the following automated and peer-reviewed criteria before merging:

- [] 1. TypeScript compilation passes with 0 errors (npm run typecheck in backend & frontend).
- [] 2. All existing and new Vitest integration tests pass (npm test in backend).
- [] 3. No merge conflicts with main; rebased on latest main prior to PR submission.
- [] 4. All new API endpoints have Zod schema validation and rate limit guards.
- [] 5. Peer review approval from at least ONE other team member and Lead Architect.

14. Top 20 Remaining Tasks Ranking (Prioritized Execution Backlog)

#	Task Title & Scope	Priority	Owner	Est. Hours	Core Justification
1	M3-01: Prisma Schema Extensions (Flashcard, Cert, etc.)	P0	Member 3	12h	Unblocks all backend and realtime feature streams.
2	M2-01: SM-2 Spaced Repetition Algorithm Engine	P0	Member 2	14h	Transforms passive quiz results into active recall.
3	M4-01: Socket.io Real-Time Room Server & Handshake	P0	Member 4	16h	Enables live peer collaboration and shared sessions.
4	M1-01: Interactive Flashcards Study Canvas UI	P0	Member 1	14h	Provides student study interface for spaced review.
5	M1-02: Virtual Study Room Layout & Synced Timer	P0	Member 1	18h	Delivers virtual study group experience.
6	M5-01: Playwright End-to-End Test Suite in CI	P1	Member 5	16h	Guarantees zero regressions across full student flows.
7	M2-03: Certificate PDF Generator & Verification	P1	Member 2	10h	Adds official credentialing for course completions.
8	M4-03: Threaded Topic Q&A Discussion Engine	P1	Member 4	12h	Enables peer discussion directly on study topics.
9	M1-03: PWA Service Worker & Offline Storage	P1	Member 1	12h	Allows study notes review without internet connectivity.
10	M2-02: Weekly Email Digest & Retention Worker	P1	Member 2	12h	Drives weekly learner re-engagement and streak reminders.
11	M5-02: Production Docker Hardening & Backup Script	P1	Member 5	12h	Secures production containers and automates DB dumps.
12	M4-02: Collaborative Whiteboard State Sync	P1	Member 4	14h	Allows synchronous note-taking in study rooms.
13	M1-04: Public Certificate Verification Page (/verify)	P1	Member 1	8h	Enables third-party credential verification.
14	M3-02: Hierarchical Comment Queries & Upvoting	P1	Member 3	12h	Ensures high performance on discussion threads.
15	M2-04: Stripe Subscription Tiers & Webhooks	P2	Member 2	14h	Enables Pro tier quota management and billing.
16	M3-03: Semantic Vector Recommendation Service	P2	Member 3	14h	Recommends content based on difficulty & history.
17	M5-03: Prometheus Alert Rules & OWASP Audit	P2	Member 5	10h	Monitors production error spikes and vulnerabilities.
18	M3-04: Test Harness MockDB Updates	P1	Member 3	6h	Maintains 100% test pass rate across new models.
19	M1-05: Discussion Forum UI Component in Notes	P1	Member 1	8h	Renders nested peer Q&A at bottom of lessons.
20	M5-04: Release Candidate Validation & Smoke Tests	P0	Member 5	8h	Final pre-launch verification across all browsers.

15. Final Project Status & Target State Vision

15.1 Current EdLearn Baseline Status

- Completed & Verified: Multi-Factor Authentication (2FA), 7 Pedagogical AI Modes, Duo Podcast Audio Synthesis, Books/Media/News Curation Hubs, Universal Bookmarks, Cmd+K Global Search, Streak Calculation Engine, 365-Day Activity Heatmap, Interactive QuizRunner, Personal Notes Scratchpad, Role-Based Access Control, Admin Portal, In-App Notifications, and AI Failover Circuit Breaker.
- Quality Baseline: 84/84 Vitest integration tests passing, 0 TypeScript compiler errors.

15.2 Target State After Completion of 5-Member Plan

- Upon completion of the 4-week implementation plan, EdLearn will stand as a state-of-the-art, fully autonomous learning platform offering:
1. Active Recall & Long-Term Mastery: Automatic conversion of quiz struggles into daily SM-2 flashcard review sessions.
 2. Synchronous Social Learning: Live peer study rooms with synchronized Pomodoro timers, shared whiteboards, and lesson Q&A.
 3. Offline Resilience: Full PWA support enabling uninterrupted note study and offline quiz caching.
 4. Certified Academic Credentials: Tamper-proof, cryptographically verifiable course completion certificates.
 5. Production-Hardened Infrastructure: Sub-200MB Docker containers, automated daily database backups, and complete Playwright E2E test coverage.

ENGINEERING MANAGEMENT APPROVAL & SIGN-OFF

Prepared by: Lead Software Architect, Product Analyst & Engineering Manager

Approved for Sprint Execution: August 20, 2026

Target Release: EdLearn v2.0 Production Release Candidate

[APPROVED FOR SPRINT EXECUTION]